Metric	DQN (100k)	Double DQN (100k)	DQN (200k total)	Double DQN (200k total)
Average Reward	2.48	2.28	7.88	**11.08**
Maximum Reward	8	8	18	**26**
Std Dev	1.69	**1.28**	**3.04**	4.06
Final Loss	0.0104	**0.0033**	0.0084	**0.0023**
Final Epsilon	0.10	0.10	0.10	0.10

During the first 100k steps, both algorithms achieved similar performance. However, Double DQN exhibited lower loss values and reduced reward variability, suggesting more stable value estimation despite comparable rewards.

What happened after the additional 100k?

DQN

Average reward: 2.48 → 7.88

Improvement: +217%
Maximum reward: 8 → 18

Double DQN

Average reward: 2.28 → 11.08

Improvement: +386%

Maximum reward: 8 → 26

At 100k: DQN ≈ Double DQN

At 200k: Double DQN clearly outperforms DQN

The benefits of Double DQN become more apparent as training progresses and value estimates become increasingly important.

Why did Double DQN's variance become higher? A higher-performing agent often experiences a wider range of rewards because it reaches more advanced game states. So increased variance is not necessarily instability. It may simply reflect a broader reward distribution at higher performance levels.

LOSS ANALYSIS

Training Stage	DQN	Double DQN
100k	0.0104	0.0033
200k	0.0084	0.0023

Double DQN generated more accurate and stable Q-value targets throughout training.

Both DQN and Double DQN demonstrated successful learning in Breakout. During the first 100k training steps, the two algorithms achieved similar reward levels, although Double DQN exhibited lower loss values and reduced reward variability. After an additional 100k training steps with a fixed exploration rate of $\varepsilon = 0.1$, Double DQN achieved substantially higher average and maximum rewards than DQN. Specifically, Double DQN reached an average reward of 11.08 compared to 7.88 for DQN, and a maximum reward of 26 compared to 18 for DQN. These results suggest that the reduction of overestimation bias in Double DQN leads to improved long-term performance.

200k Steps

- DQN: Avg Reward = 7.88
- Double DQN: Avg Reward = 11.08
- Double DQN outperformed DQN by ~41%

Double DQN showed little advantage early in training, but achieved significantly higher rewards during extended training, supporting the hypothesis that reducing overestimation bias improves long-term learning performance.

Sample Efficiency: How much reward was obtained per interaction with the environment?

Under an equal interaction budget, Double DQN achieved higher performance, indicating superior sample efficiency during extended training.

Metric	**DQN (200k)**	**Double DQN (200k)**
Avg Reward	7.88	11.08
Max Reward	18	26
Std Dev	3.04	4.06
Final Loss	0.0084	0.0023